Java Architecture — JDK, JRE and JVM

How Java source code becomes a platform-independent application

Java Source Code

Example: HelloWorld.java

javac compiler

JDK — Java Development Kit

Development tools:

- javac — Java compiler
- java — application launcher
- javadoc — documentation tool
- Debugging, packaging and other development utilities

compiled bytecode

JRE — Java Runtime Environment

Provides the environment required to run Java applications

JVM — Java Virtual Machine

Loads, verifies and executes Java bytecode

machine-level execution

Operating System & Hardware

Windows • Linux • macOS • Other supported platforms

The JVM provides the platform-specific implementation needed to run the bytecode.

Platform Independence

Java source → bytecode → JVM → platform-specific execution

The same .class bytecode can run on different operating systems when a compatible JVM is available.